

Problem

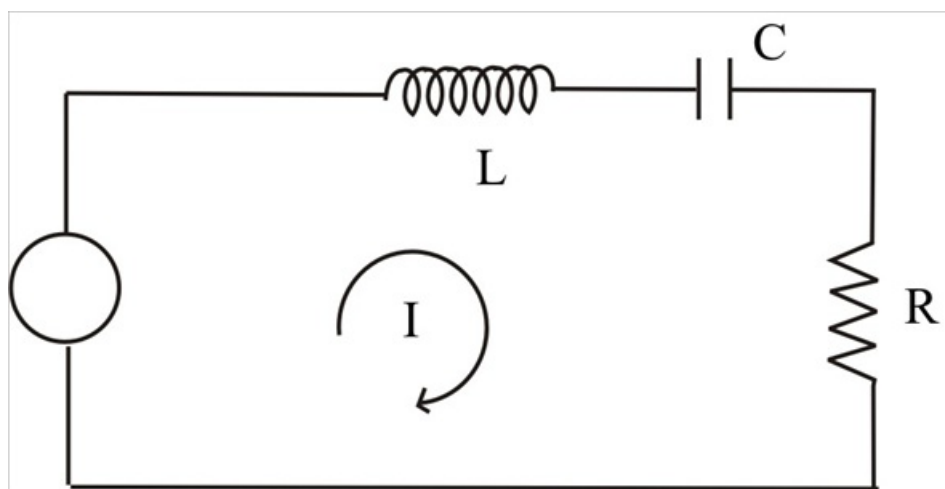
Determine the range of frequencies that will be passed by a series RLC band-pass filter with $R = 10\ \Omega$, $L = 25\ \text{mH}$, and $C = 0.4\ \mu\text{F}$. Find the quality factor.

Step-by-step solution

Step 1 of 5

Consider RLC Band width pass filter

Step 2 of 5



Step 3 of 5

Transfer function is $H(\omega) = \frac{V_o}{V_i}$ —(1)

$V_o \rightarrow$ Out put voltage

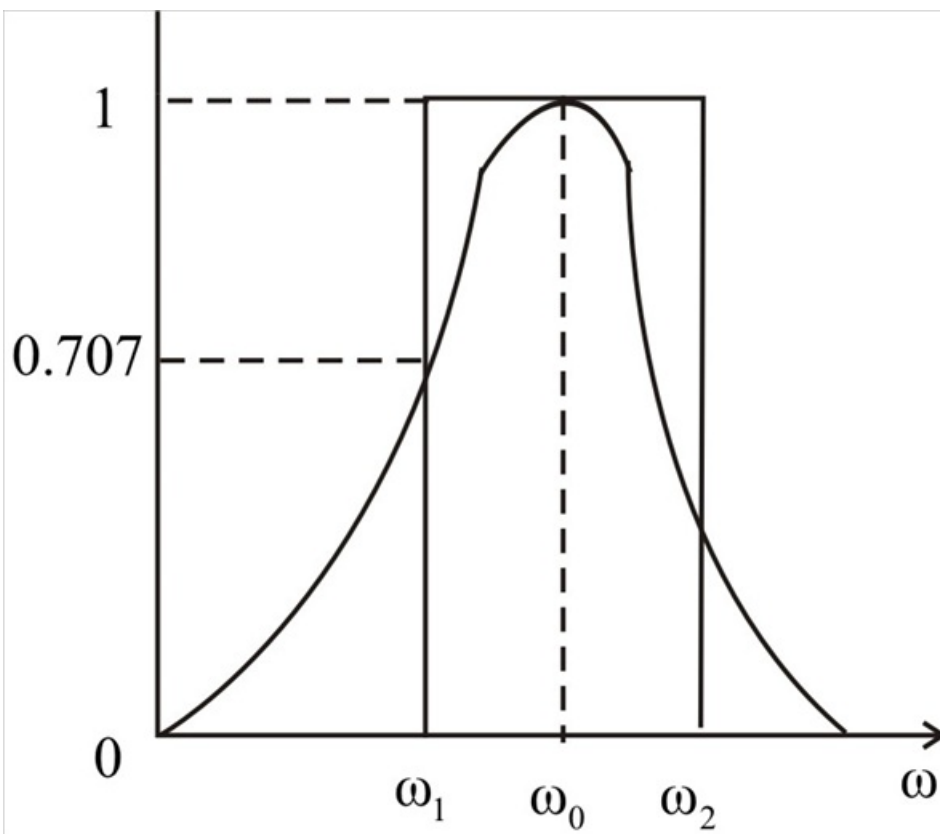
$V_i \rightarrow$ In put voltage

or $H(\omega) = \frac{R}{R + j\left(\omega L - \frac{1}{\omega C}\right)}$ —(2)

$H(0) = 0 \quad H(\infty) = 0$

$(\omega_1 < \omega < \omega_2)$ —(3)

Step 4 of 5



Step 5 of 5

We know the central frequency

$$\omega_o = \frac{1}{\sqrt{LC}} \quad -(4)$$

and Band width $B = \frac{R}{L} = \frac{\omega_o}{Q} \quad -(5)$

and Quality factor, $Q = \frac{\omega_o L}{R} = \frac{1}{\omega_o CR} \quad -(6)$

Step, we have given

$$R = 10\Omega, \quad L = 25MH, \quad C = 0.4\mu F.$$

now from (5)

$$B = \frac{10}{25 \times 10^{-3}} = 400 \text{ rad/s}$$

$$\text{and, } \omega_o = \frac{1}{\sqrt{25 \times 10^{-3} \times 0.4 \times 10^{-6}}} = 10^4 \text{ rad/s}$$

$$\text{form (6)} \quad Q = \frac{\omega_o L}{R} = \frac{10^4 \times 25 \times 10^{-3}}{10} = 25$$

$$\boxed{Q = 25}$$

as $Q \geq 10$, then,

$$\omega_1 = \omega_o - \frac{B}{2} \quad -(7)$$

$$\text{and } \omega_2 = \omega_o + \frac{B}{2} \quad -(8)$$

$$\text{from (7)} \quad \omega_1 = 10000 - 200 = 9800$$

$$\omega_1 = 2\pi f_1 \Rightarrow f_1 = \frac{9800}{2\pi} = 1.56 \text{ KHz}$$

$$\omega_2 = 10200 \Rightarrow f_2 = \frac{10200}{2\pi} = 1.62 \text{ KHz}$$

$$\boxed{f_1 = 1.56 \text{ KHz}}$$

$$\boxed{f_2 = 1.62 \text{ KHz}}$$